

FORMULA SHEET · 01

PCL-AI Formula Sheet

PCI AI Project Controls Leader — the complete quantitative reference

CPI · SPI · EAC · TCPI · ES · TF · EMV · CCC

PCI AI · PROJECT CONTROLS INSTITUTE GLOBAL

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AI proposes. The professional disposes.

How to use this sheet

Every formula the PCL-AI Body of Knowledge teaches, in one place, with the Knowledge Area where it is worked in full.

Three columns matter. **Formula** is the expression. **Meaning** is what it answers. **Watch for** is the mistake that formula actually attracts in practice — the reason a correct expression still produces a wrong answer.

This sheet is a study aid. It is **not provided in the examination**; candidates are expected to know the core formulas.

Conventions. Currency in USD unless stated. Ratios and indices to 2 dp. Adverse amounts in parentheses. Periods are consistent within any single calculation — never mix monthly and cumulative figures in one expression.

The notation clash rule. **PV** means **Planned Value** in earned-value contexts and **present value** in discounting contexts. Write "present value" in words, or **PV(x)**, whenever discounting. These two collide on the same page more often than any other pair of symbols in the discipline.

1. Notation

SYMBOL	MEANING	UNIT
A , L , E	Assets, liabilities, equity	currency
PV (BCWS)	Planned value / budgeted cost of work scheduled	currency
EV (BCWP)	Earned value / budgeted cost of work performed	currency
AC (ACWP)	Actual cost / actual cost of work performed	currency
BAC	Budget at completion	currency
CV , SV	Cost variance, schedule variance	currency
CPI , SPI	Cost performance index, schedule performance index	ratio
EAC , ETC	Estimate at completion, estimate to complete	currency
VAC	Variance at completion	currency
TCPI	To-complete performance index	ratio
ES , AT	Earned schedule, actual time	periods
PoC	Percentage of completion	%
O , M , P	Optimistic, most likely, pessimistic duration	periods
tE , σ	PERT expected duration, standard deviation	periods
ES , EF , LS , LF	Early start, early finish, late start, late finish	date/period
TF , FF	Total float, free float	periods
r , n	Discount rate per period, number of periods	ratio, count
EMV	Expected monetary value	currency

2. Accounting and financial reporting · Domains 1-2

FORMULA	MEANING	WATCH FOR	KA
$A = L + E$	The accounting equation	—	1.1.1
$\Sigma \text{ Debits} = \Sigma \text{ Credits}$	Double-entry invariant	A trial balance that balances proves arithmetic, not correctness	1.1.3
$\text{Retained earnings} = \text{Opening} + \text{Income} - \text{Expenses} - \text{Distributions}$	Equity movement	—	1.1.1
$\text{Depreciation} = (\text{Cost} - \text{Residual}) \div \text{Useful life}$	Straight line	Residual omitted; life not reassessed	1.3.4
$\text{Carrying amount} = \text{Cost} - \text{Accumulated depreciation} - \text{Impairment}$	Net book value	—	1.3.4
$\text{Expected value} = \Sigma (\text{probability} \times \text{outcome})$	Provisions, variable consideration	Using most-likely outcome where the population is large	1.4.3 / 2.2.4
$\text{Present value} = \text{Amount} \div (1 + r)^n$	Discounting a provision	—	1.4.3
$\text{PoC} = \text{Costs incurred} \div \text{Total estimated costs}$	Cost-to-cost percentage of completion	Uninstalled materials inflating the numerator	2.2.6
$\text{Cumulative revenue} = \text{PoC} \times \text{Transaction price}$	IFRS 15 over-time revenue	Applying to cumulative, then forgetting to deduct revenue already recognised	2.2.6
$\text{Allocated price}_i = \text{Transaction price} \times (\text{SSP}_i \div \Sigma \text{ SSP})$	IFRS 15 allocation	—	2.2.5
$\text{Capitalised borrowing cost} = \text{Weighted-avg qualifying expenditure} \times \text{rate}$	IAS 23	Capitalisation continued through a suspension	2.4.4
$\text{Contract asset} / (\text{liability}) = \text{Revenue recognised} - \text{Amounts billed}$	Over/under-billing position	Sign convention reversed	7.5

3. Budgeting, cost and performance · Domains 3-5

FORMULA	MEANING	WATCH FOR	KA
$BAC = \Sigma \text{ control-account budgets} + \text{contingency}$	Cost baseline	Management reserve wrongly included	3.1.4
$\text{Total authorised budget} = BAC + \text{Management reserve}$	Full authorisation	—	3.1.4
$\text{Analogous estimate} = \text{Past cost} \times (\text{this driver} \div \text{past driver})$	Analogous estimating	Driver not actually the cost driver	3.2.2
$\text{Parametric estimate} = \text{Parameter} \times \text{Rate}$	Parametric estimating	Rate from a different scope basis	3.2.2
$\text{Total cost} = \text{Fixed} + (\text{Variable per unit} \times \text{Volume})$	Cost behaviour	—	5.1.1
$OAR = \text{Budgeted overhead} \div \text{Budgeted activity base}$	Overhead absorption rate	—	5.1.3
$\text{Over/(under) absorption} = \text{Absorbed} - \text{Incurred}$	Absorption variance	—	5.1.3
$\text{Cost to date} = \text{Actuals} + \text{Accruals}$	True cost to date	The single most damaging omission in project controls	5.2.1
$\text{Price variance} = (\text{Actual price} - \text{Standard price}) \times \text{Actual quantity}$	Variance decomposition	Standard quantity used in place of actual	4.2.3
$\text{Quantity variance} = (\text{Actual qty} - \text{Standard qty}) \times \text{Standard price}$	Variance decomposition	Actual price used in place of standard	4.2.3
$\text{Total variance} = \text{Price variance} + \text{Quantity variance}$	Reconciliation check	If the two do not sum to the total, one used the wrong multiplier	4.2.3

The accrual rule. Every index below depends on **AC**. If **AC** excludes work performed but not yet invoiced, **CPI** overstates performance and **EAC** understates cost — and both errors are invisible in the report. Reconcile goods-received-not-invoiced to booked accruals before computing anything in Section 4.

4. Earned value and forecasting · Domains 6, 9

4.1 POSITION

FORMULA	MEANING	WATCH FOR	KA
$CV = EV - AC$	Cost variance	Negative is adverse	6.2.1
$SV = EV - PV$	Schedule variance (in currency)	Goes to zero at completion regardless of lateness	6.2.1
$CPI = EV \div AC$	Cost efficiency to date	Computed on invoiced cost only	6.2.2
$SPI = EV \div PV$	Schedule efficiency in cost terms	Also returns to 1.00 at completion	6.2.2
$\% \text{ complete} = EV \div BAC$	Progress by value	Not the same as physical or time progress	6.1

4.2 FORECAST

FORMULA	ASSUMPTION IT ENCODES	KA
$EAC = AC + ETC$	Identity — always true	6.3.1
$EAC = AC + (BAC - EV)$	The variance was a one-off ; remaining work performs to plan	6.3.2
$EAC = BAC \div CPI$	Cost performance to date persists to completion	6.3.2
$EAC = AC + (BAC - EV) \div CPI$	Same assumption, stated in remaining-work form	6.3.2
$EAC = AC + (BAC - EV) \div (CPI \times SPI)$	Both cost and schedule pressure persist	6.3.2
$VAC = BAC - EAC$	Variance at completion; negative is an overrun	6.3.4
$TCPI \text{ (to BAC)} = (BAC - EV) \div (BAC - AC)$	Efficiency required to still hit the budget	6.2.3
$TCPI \text{ (to EAC)} = (BAC - EV) \div (EAC - AC)$	Efficiency required to hit a revised forecast	6.2.3

Identity worth knowing. $BAC \div CPI$ is algebraically identical to $AC \times BAC \div EV$. And when $EAC = BAC \div CPI$, then $TCPI \text{ (to EAC)} = CPI$ exactly — the forecast assumes you continue performing precisely as you have been. If a forecast produces a TCPI far above the CPI achieved to date, it is asserting an improvement nobody has yet demonstrated.

4.3 EARNED SCHEDULE

FORMULA	MEANING	WATCH FOR	KA
$ES = M + (EV - PV_M) \div (PV_{M+1} - PV_M)$	Interpolate between the cumulative-PV periods bracketing EV	Using period rather than cumulative PV	6.4.3
$SV(t) = ES - AT$	Schedule variance in time	—	6.4.3
$SPI(t) = ES \div AT$	Time-based schedule efficiency	—	6.4.3

Why earned schedule exists. SV and SPI are computed in currency and both converge to zero and 1.00 at completion, however late the project finishes. $SV(t)$ and $SPI(t)$ do not, which makes them the more honest late-stage measures.

4.4 ADAPTIVE DELIVERY

FORMULA	MEANING	WATCH FOR	KA
$\% \text{ complete} = \text{Points completed} \div \text{Total planned points}$	AgileEVM progress	Point inflation across sprints	9.5.3
$EV = \% \text{ complete} \times BAC$	Earned value in adaptive delivery	—	9.5.3
$Velocity = \text{Points completed} \div \text{Sprint}$	Delivery rate	Averaging across a changed team	9.3
$\text{Sprints remaining} = \text{Points remaining} \div \text{Velocity}$	Forecast to completion	Using best-ever velocity rather than rolling average	9.3
$\text{Run rate} = \text{Team cost} \div \text{Sprint}$	Cost per sprint	—	9.5.2
$\text{Cycle time} = WIP \div \text{Throughput}$	Little's Law	Only valid for a stable system over the measured period	9.4

5. Scheduling and risk · Domains 10, 12

FORMULA	MEANING	WATCH FOR	KA
$EF = ES + \text{Duration}$	Forward pass	Off-by-one at period boundaries	10.2
$LS = LF - \text{Duration}$	Backward pass	—	10.2
$TF = LS - ES = LF - EF$	Total float	If the two expressions disagree, the pass is wrong	10.2.4
$FF = \min(\text{successor } ES) - EF$	Free float	Ignoring lags on the successor link	10.2.4
Critical path: $TF = 0$	Longest path	Constraints creating artificial zero float	10.2.3
$tE = (O + 4M + P) \div 6$	PERT expected duration	Three points sourced from one person	10.1.4
$\sigma = (P - O) \div 6$	PERT standard deviation	—	10.1.4
$\sigma^2 \text{ path} = \sum \sigma^2 \text{ of activities on the path}$	Path variance	Variances add; standard deviations do not	10.3.4
$\text{Crash cost slope} = (\text{Crash cost} - \text{Normal cost}) \div (\text{Normal duration} - \text{Crash duration})$	Cost per period saved	Crashing an activity that is not on the critical path	10.3.1
$EMV = \text{Probability} \times \text{Impact}$	Expected monetary value	Applied to a one-off event where the expected value never occurs	12.2.3
$\text{Contingency} \approx \sum EMV \text{ (or P80 - P50 from simulation)}$	Risk-based contingency	Percentage applied in place of quantification	12.3.1

Variance adds, deviation does not. To combine schedule uncertainty along a path, sum the **variances** (σ^2), then take the square root. Summing standard deviations overstates path uncertainty substantially.

6. Commercial · Domain 7

FORMULA	MEANING	WATCH FOR	KA
$\text{Fee} = \text{Target fee} + \text{Share ratio} \times (\text{Target cost} - \text{Actual cost})$	CPIF incentive fee	Sign convention on an overrun	7.1.3
$\text{Pain/gain} = \text{Share ratio} \times (\text{Actual} - \text{Target})$	Target-cost mechanism	—	7.1.4
$\text{LD exposure} = \text{LD rate} \times \text{Days late}$	Liquidated damages	Cap not applied	7.2.3
$\text{Amount due} = \sum (\% \text{ complete} \times \text{item amount}) - \text{Retention} - \text{Previous payments}$	Interim payment application	Retention release omitted at the milestone	7.4.3
$\text{Retention released} = \text{Retention held} \times \text{Release \%}$	Retention mechanics	—	7.4.4
$\text{Remeasured value} = \text{Actual quantity} \times \text{BoQ rate}$	Remeasurement contract	Rate applied to the wrong measured unit	7.3.4

7. Process cycles and working capital · Domain 11

FORMULA	MEANING	WATCH FOR	KA
$DSO = \text{Receivables} \div \text{Daily revenue}$	Days sales outstanding	Revenue not annualised consistently	11.1.3
$DIO = \text{Inventory} \div \text{Daily COGS}$	Days inventory outstanding	—	11.A.1
$DPO = \text{Payables} \div \text{Daily COGS}$	Days payables outstanding	—	11.A.1
$CCC = DSO + DIO - DPO$	Cash conversion cycle	Sign on DPO	11.A.1
$\text{Cash released} \approx \Delta DSO \times \text{Daily revenue}$	Working capital freed	—	11.A.1

8. Decision aid — which EAC method

The formula is not the difficulty. Selecting and defending the assumption is.

QUESTION	IF YES	METHOD
Was the variance caused by a discrete, closed event that will not recur?	→	$EAC = AC + (BAC - EV)$
Does the remaining work resemble the work done, with performance likely to persist?	→	$EAC = BAC \div CPI$
Is the project also behind, with schedule pressure driving cost?	→	$EAC = AC + (BAC - EV) \div (CPI \times SPI)$
Is the remaining scope materially different from what has been done?	→	Bottom-up ETC; indices do not apply

Publish the range, not the point. Report at least two scenarios with the assumption named against each. A single EAC quoted to the nearest thousand implies a precision the method cannot support.

9. Decision aid — which progress measurement method

WORK PACKAGE CHARACTER	METHOD	NOTE
Short duration, spans one period	0/100	No partial credit; simplest and least gameable
Two periods, discrete start and finish	50/50	Credit on start and completion
Countable, homogeneous output	Units complete	Strongest objective measure
Long duration, distinct verifiable stages	Milestone weighting	Weights fixed before work starts
Level of effort, no discrete output	Apportioned / LoE	Earns with time; never the default
Cannot be measured physically	—	Re-scope the work package

The rule. Fix the measurement method per work package before work begins, and never accept a percentage supplied by the party being measured. That figure is a commercial position, not a measurement.

10. Worked micro-examples

Each is internally consistent and can be reproduced from the sheet.

Earned value at a data date. PV 168,000 · EV 151,200 · AC invoiced 158,000 · accrual 12,400 · BAC 420,000.

- True AC = 158,000 + 12,400 = **170,400**
- CPI = 151,200 ÷ 170,400 = **0.887** (on invoiced cost alone it would read 0.957)
- SPI = 151,200 ÷ 168,000 = **0.900**
- EAC one-off = 170,400 + (420,000 – 151,200) = **439,200**
- EAC CPI persists = 420,000 ÷ 0.887 = **473,300**
- EAC both persist = 170,400 + 268,800 ÷ (0.887 × 0.900) = **507,000**
- VAC range = **(19,200) to (87,000)**

The accrual alone moves the forecast by 34,400 — the difference between a contained overrun and an uncontained one.

Earned schedule. Cumulative PV by month: 20, 55, 100, 150, 210, 280. EV = 170 at AT = 5.

- EV falls between month 4 (150) and month 5 (210)
- ES = 4 + (170 – 150) ÷ (210 – 150) = 4 + 0.333 = **4.333**
- $SV(t)$ = 4.333 – 5 = **(0.667) months**
- $SPI(t)$ = 4.333 ÷ 5 = **0.867**

PERT. O = 8, M = 12, P = 22.

- tE = (8 + 48 + 22) ÷ 6 = **13.0**
- σ = (22 – 8) ÷ 6 = **2.33**; σ^2 = **5.44**

Float. ES = 4, duration 6, LF = 14.

- EF = 10 · LS = 8 · TF = 8 – 4 = **4** = 14 – 10 — the two forms agree

Working capital. Receivables 4,500,000 · daily revenue 50,000 · inventory 2,000,000 · daily COGS 40,000 · payables 3,200,000.

- DSO = **90** · DIO = **50** · DPO = **80** · CCC = 90 + 50 – 80 = **60 days**

11. The ten formulas most often misapplied

1. **CPI computed on invoiced cost.** Without accruals, every downstream index and forecast is wrong in the same direction.
2. **SV read as a schedule measure late in a project.** It is denominated in currency and converges to zero at completion regardless of lateness. Use $SV(t)$.
3. **A single EAC presented as the forecast.** Three defensible answers exist. The range and the named assumption are the deliverable.
4. **$TCPI$ ignored as a reality check.** A $TCPI$ well above the CPI achieved to date asserts an unearned improvement.
5. **Percentage complete supplied by the measured party.** A commercial position presented as a measurement.
6. **Standard deviations summed along a schedule path.** Variances add; take the square root at the end.
7. **EMV applied to a single one-off event.** Expected value describes a portfolio of risks, not a binary outcome that will either happen or not.
8. **Contingency set as a percentage of budget.** Unless it reconciles to quantified risk exposure, it is a habit rather than an analysis.

9. **Crashing an activity off the critical path.** Cost incurred, zero duration saved.

10. **Management reserve included in BAC . BAC** is the cost baseline; management reserve sits outside it and is released by governance.

12. Cross-references

FOR THE FULL TREATMENT	SEE
Every formula worked with figures and self-check exercises	The PCL-AI Body of Knowledge, at the KA cited in each row
The published syllabus and weightings	PCL-AI Examination Content Outline
Why cut-off and accrual discipline dominate outcomes	Knowing Early, Knowledge Series 01, §4 and §16
Assessing whether a professional can defend these methods	Show Your Work, Knowledge Series 02, §4 and §8

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